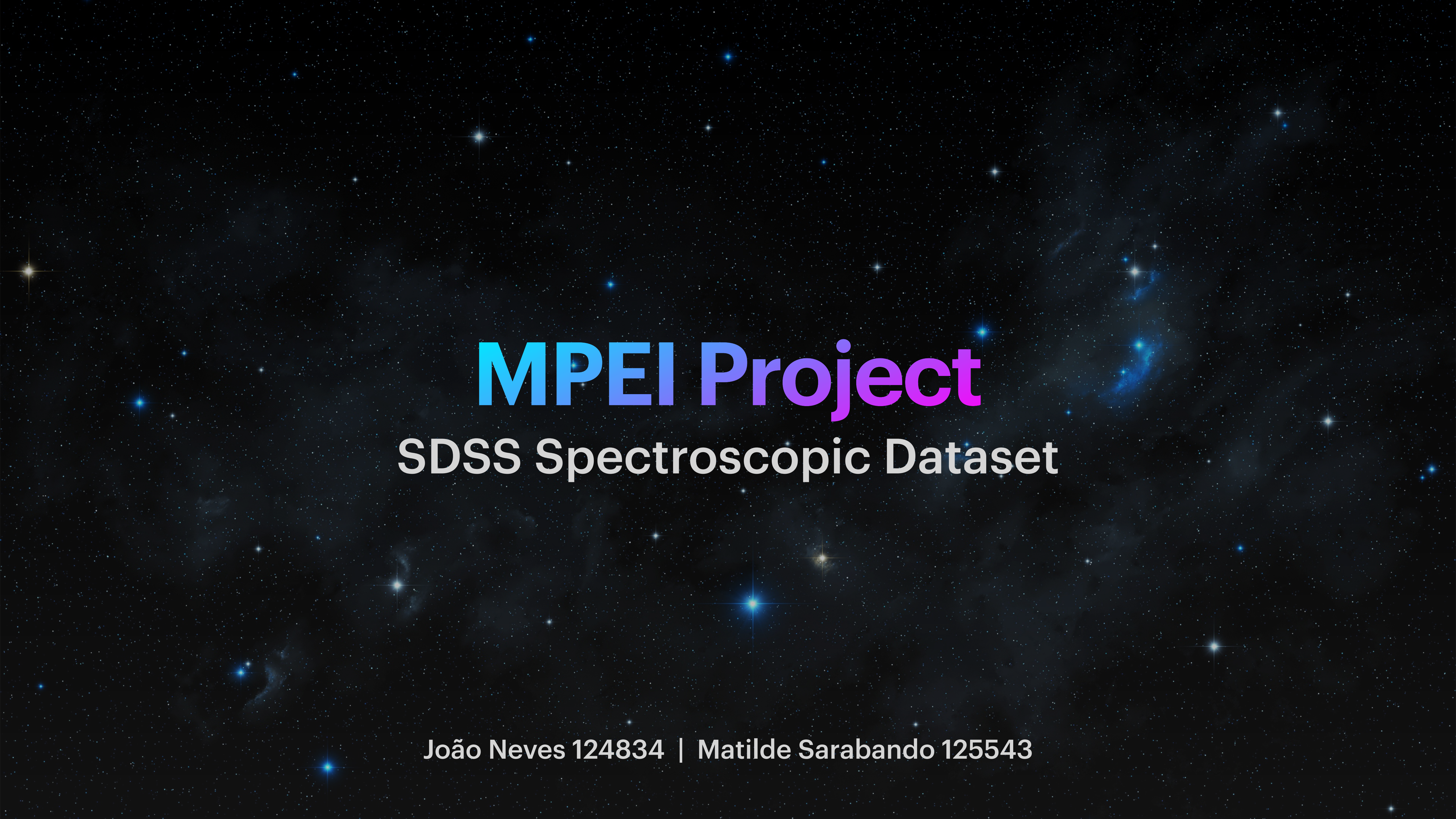




MPEI Project

SDSS Spectroscopic Dataset

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Project overview

Dataset from the Sloan Digital Sky Survey

- Our project is about celestial objects and uses as a base a dataset from the SDSS
- The final application should be able to use:
 - **Naïve Bayes** algorithm to classify the different objects in the dataset into some categories based on their data
 - **Bloom Filters** to create sets of some of those instances and check if a specific instance belongs or not to that specific set
 - **Minhash** to check similarities between the instances of the dataset according to their available data on the dataset

The dataset

Structure and what it contains

- Some additional information about our dataset can be consulted [here](#)
- The dataset contains objects (one per line) classified in 3 different types of celestial bodies with some subclasses each:
 - Galaxy
 - Starforming, Starburst or AGN (Active Galaxy Nuclei)
 - Star
 - O, B, A, F, G, K, M, Carbon or CarbonWD (white dwarf), CV (Cataclysmic Variables)
 - QSO (quasar)
 - Broadline

The dataset

Columns

- The dataset has 194 columns, and these are the ones we are using to identify and define the objects:
 - **Target Variable:**
 - **class** - label to predict Star, Galaxy, or Quasar
 - **Physical Features:**
 - **z** - redshift
 - **velDisp** - velocity dispersion
 - **Quality/Precision Feature:**
 - **snMedian_r** - median signal-to-noise ratio in the red band
 - **Spectral Flux Features:**
 - **spectroFlux_u, spectroFlux_g, spectroFlux_r, spectroFlux_i, spectroFlux_z** - the frequency emitted by the object in the different bands of the spectrum (ultraviolet, green/blue, red, far red/near-infrared and near-infrared, respectively)

Naïve Bayes

Implemented

- We use the Naïve Bayes algorithm to **map each object into the most probable category** among the ones explained earlier, based on the parameters shown in the last slide
- The algorithm uses the **redshift** as the primary discriminator, because it differs substantially from stars (very low), to galaxies (moderate) and quasars (very high)
- Then it uses the Pogson's logarithmic scale formula to map the spectral fluxes to **4 discrete color index features**, whose combinations can help identify with some accuracy one of the 3 categories (eg. quasar with lower redshifts has lower u-g)
- **Velocity dispersion** and **signal-to-noise ratio** are also used as distinctive parameters
- The combination of these parameters can create kind of a **signature** of values for each type

Bloom Filter

Implemented

- The Bloom Filter implementation is used with the 3 categories and the Naïve Bayes algorithm
- For the Bloom Filter logic, because we didn't have a parameter that was unique for each entry (line), we mapped each of them to a unique sequential ID, corresponding to the position in the dataset fragment from 0 to $n-1$, where n was the number of objects in that fragment
- We used that ID in the hash functions (created according to the Carter-Wegman method) to save the desired number of hashes (k) for each entry in the bloom filter array
- We use the same hash to later verify if an element with a specific ID belongs or not to the filter (considered set of objects)
- For the Carter-Wegman method, the parameters a and b are randomly generated according to the chosen k , the number of hashes per entry on the bloom filter

Minhash

In progress

- Minhash will be used in our project to check the similarity between the different objects according to their values in their entries in the dataset
- Because most of the data is **continuous**, we need to discretize it first before we can use it to feed the Minhash method
- Regarding this, we are still discussing about the best way to implement it
- The main idea would be to break the universe of possible values in a considerable number of discrete intervals and map each value to one of these discrete intervals
- This way we could apply the Minhash algorithm to this discretized data and, using the values of the relevant fields in Naïve Bayes, determine similar objects inside the dataset